43. CLEBSCH-GORDAN COEFFICIENTS, SPHERICAL HARMONICS, AND *d* FUNCTIONS

Note: A square-root sign is to be understood over *every* coefficient, *e.g.*, for $-8/15$ read $-\sqrt{8/15}$.

Notation:

J	J	...
M	M	...

m_1	m_2	
m_1	m_2	Coefficients
.	.	
.	.	
.	.	

1/2×1/2

	1		
+1/2	1	1	0
+1/2	1	0	0
+1/2	−1/2	1/2	1/2
−1/2	+1/2	1/2	−1/2
	−1/2	−1/2	1

1×1/2

	3/2			
+1	+3/2	3/2	1/2	
+1	+1/2	+1/2	+1/2	
+1	−1/2	1/3	2/3	3/2
0	+1/2	2/3	−1/3	−1/2
0	−1/2	2/3	1/3	3/2
−1	+1/2	1/3	−2/3	−3/2
	−1	−1/2		1

2×1/2

	5/2			
+2	+5/2	5/2	3/2	
+2	+1/2	1	+3/2	+3/2
+2	−1/2	1/5	4/5	5/2	3/2
+1	+1/2	4/5	−1/5	+1/2	+1/2
+1	−1/2	2/5	3/5	5/2	3/2
0	+1/2	3/5	−2/5	−1/2	−1/2
0	−1/2	3/5	2/5	5/2	3/2
−1	+1/2	2/5	−3/5	−3/2	−3/2
−1	−1/2	4/5	1/5	5/2	
−2	+1/2	1/5	−4/5	−5/2	
	−2	−1/2		1	

2×1

	3			
+2	+3	3	2	
+2	+1	1	+2	+2
+2	0	1/3	2/3	3	2	1
+1	+1	2/3	−1/3	+1	+1	+1
+2	−1	1/15	1/3	3/5			
+1	0	8/15	1/6	−3/10	3	2	1
0	+1	2/5	−1/2	1/10	0	0	0
+1	−1	1/5	1/2	3/10			
0	0	3/5	0	−2/5	3	2	1
−1	+1	1/5	−1/2	3/10	−1	−1	−1
0	−1	2/5	1/2	1/10			
−1	0	8/15	−1/6	−3/10	3	2	
−2	+1	1/15	−1/3	3/5	−2	−2	
−1	−1	2/3	1/3	3			
−2	0	1/3	−2/3	−3			
−2	−1	1					

3/2×1/2

	2			
+2	+2	2	1	
+3/2	+1/2	1	+1	+1
+3/2	−1/2	1/4	3/4	2	1
+1/2	+1/2	3/4	−1/4	0	0
+1/2	−1/2	1/2	1/2	2	1
−1/2	+1/2	1/2	−1/2	−1	−1
−1/2	−1/2	3/4	1/4	2	
−3/2	+1/2	1/4	−3/4	−2	
−3/2	−1/2	1			

1×1

	2			
+1	+2	2	1	
+1	+1	1	+1	+1
+1	0	1/2	1/2	2	1	0
0	+1	1/2	−1/2	0	0	0
+1	−1	1/6	1/2	1/3			
0	0	2/3	0	−1/3	2	1
−1	+1	1/6	−1/2	1/3	−1	−1
0	−1	1/2	1/2	2			
−1	0	1/2	−1/2	−2			
−1	−1	1					

3/2×1

	5/2			
+3/2	+5/2	5/2	3/2	
+3/2	+1	1	+3/2	+3/2
+3/2	0	2/5	3/5	5/2	3/2	1/2
+1/2	+1	3/5	−2/5	+1/2	+1/2	+1/2
+3/2	−1	1/10	2/5	1/2			
+1/2	0	3/5	1/15	−1/3	5/2	3/2	1/2
−1/2	+1	3/10	−8/15	1/6	−1/2	−1/2	−1/2
+1/2	−1	3/10	8/15	1/6			
−1/2	0	3/5	−1/15	−1/3	5/2	3/2	
−3/2	+1	1/10	−2/5	1/2	−3/2	−3/2	
−1/2	−1	3/5	2/5	5/2			
−3/2	0	2/5	−3/5	−5/2			
−3/2	−1	1					

3/2×3/2

	3			
+3/2	+3	3	2	
+3/2	+3/2	1	+2	+2
+3/2	+1/2	1/2	1/2	3	2	1
+1/2	+3/2	1/2	−1/2	+1	+1	+1
+3/2	−1/2	1/5	1/2	3/10			
+1/2	+1/2	3/5	0	−2/5	3	2	1	0
−1/2	+3/2	1/5	−1/2	3/10	0	0	0	0
+3/2	−3/2	1/20	1/4	9/20	1/4			
+1/2	−1/2	9/20	1/4	−1/20	−1/4			
−1/2	+1/2	9/20	−1/4	−1/20	1/4			
−3/2	+3/2	1/20	−1/4	9/20	−1/4	3	2	1
+1/2	−3/2	1/5	1/2	3/10		−1	−1	−1
−1/2	−1/2	3/5	0	−2/5	3	2		
−3/2	+1/2	1/5	−1/2	3/10	−2	−2		
−1/2	−3/2	1/2	1/2	3				
−3/2	−1/2	1/2	−1/2	−3				
−3/2	−3/2	1						

2×3/2

	7/2				
+2	+7/2	7/2	5/2		
+2	+3/2	1	+5/2	+5/2	
+2	+1/2	3/7	4/7	7/2	5/2	3/2
+1	+3/2	4/7	−3/7	+3/2	+3/2	+3/2
+2	−1/2	1/7	16/35	2/5			
+1	+1/2	4/7	1/35	−2/5	7/2	5/2	3/2	1/2
0	+3/2	2/7	−18/35	1/5	+1/2	+1/2	+1/2	+1/2
+2	−3/2	1/35	6/35	2/5	2/5				
+1	−1/2	12/35	5/14	0	−3/10				
0	+1/2	18/35	−3/35	−1/5	1/5	7/2	5/2	3/2	1/2
−1	+3/2	4/35	−27/70	2/5	−1/10	−1/2	−1/2	−1/2	−1/2
+1	−3/2	4/35	27/70	2/5	1/10				
0	−1/2	18/35	3/35	−1/5	−1/5				
−1	+1/2	12/35	−5/14	0	3/10	7/2	5/2	3/2	
−2	+3/2	1/35	−6/35	2/5	−2/5	−3/2	−3/2	−3/2	
0	−3/2	2/7	18/35	1/5					
−1	−1/2	4/7	−1/35	−2/5	7/2	5/2			
−2	+1/2	1/7	−16/35	2/5	−5/2	−5/2			
−1	−3/2	4/7	3/7	7/2					
−2	−1/2	3/7	−4/7	−7/2					
−2	−3/2	1							

2×2

	4			
+2	+4	4	3	
+2	+2	1	+3	+3
+2	+1	1/2	1/2	4	3	2
+1	+2	1/2	−1/2	+2	+2	+2
+2	0	3/14	1/2	2/7				
+1	+1	4/7	0	−3/7	4	3	2	1
0	+2	3/14	−1/2	2/7	+1	+1	+1	+1
+2	−1	1/14	3/10	3/7	1/5				
+1	0	3/7	1/5	−1/14	−3/10				
0	+1	3/7	−1/5	−1/14	3/10				
−1	+2	1/14	−3/10	3/7	−1/5	4	3	2	1	0
+2	−2	1/70	1/10	2/7	2/5	1/5				
+1	−1	8/35	2/5	1/14	−1/10	−1/5				
0	0	18/35	0	−2/7	0	1/5				
−1	+1	8/35	−2/5	1/14	1/10	−1/5	4	3	2	1
−2	+2	1/70	−1/10	2/7	−2/5	1/5	−1	−1	−1	−1
+1	−2	1/14	3/10	3/7	1/5					
0	−1	3/7	1/5	−1/14	−3/10					
−1	0	3/7	−1/5	−1/14	3/10					
−2	+1	1/14	−3/10	3/7	−1/5	4	3	2		
0	−2	3/14	1/2	2/7		−2	−2	−2		
−1	−1	4/7	0	−3/7	4	3				
−2	0	3/14	−1/2	2/7	−3	−3				
−1	−2	1/2	1/2	4						
−2	−1	1/2	−1/2	−4						
−2	−2	1								

$$Y_1^0 = \sqrt{\frac{3}{4\pi}}\cos\theta$$

$$Y_1^1 = -\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{i\phi}$$

$$Y_2^0 = \sqrt{\frac{5}{4\pi}}\left(\frac{3}{2}\cos^2\theta - \frac{1}{2}\right)$$

$$Y_2^1 = -\sqrt{\frac{15}{8\pi}}\sin\theta\cos\theta\, e^{i\phi}$$

$$Y_2^2 = \frac{1}{4}\sqrt{\frac{15}{2\pi}}\sin^2\theta\, e^{2i\phi}$$

$$Y_\ell^{-m} = (-1)^m Y_\ell^{m*}$$

$$d^\ell_{m,0} = \sqrt{\frac{4\pi}{2\ell+1}}\, Y_\ell^m\, e^{-im\phi}$$

$$\langle j_1 j_2 m_1 m_2 | j_1 j_2 J M\rangle = (-1)^{J-j_1-j_2}\langle j_2 j_1 m_2 m_1 | j_2 j_1 J M\rangle$$

$$d^j_{m',m} = (-1)^{m-m'} d^j_{m,m'} = d^j_{-m,-m'}$$

$$d^1_{0,0} = \cos\theta \qquad d^{1/2}_{1/2,1/2} = \cos\frac{\theta}{2} \qquad d^1_{1,1} = \frac{1+\cos\theta}{2}$$

$$d^{1/2}_{1/2,-1/2} = -\sin\frac{\theta}{2} \qquad d^1_{1,0} = -\frac{\sin\theta}{\sqrt{2}}$$

$$d^1_{1,-1} = \frac{1-\cos\theta}{2}$$

$$d^{3/2}_{3/2,3/2} = \frac{1+\cos\theta}{2}\cos\frac{\theta}{2}$$

$$d^{3/2}_{3/2,1/2} = -\sqrt{3}\frac{1+\cos\theta}{2}\sin\frac{\theta}{2} \qquad d^2_{2,2} = \left(\frac{1+\cos\theta}{2}\right)^2$$

$$d^{3/2}_{3/2,-1/2} = \sqrt{3}\frac{1-\cos\theta}{2}\cos\frac{\theta}{2} \qquad d^2_{2,1} = -\frac{1+\cos\theta}{2}\sin\theta$$

$$d^{3/2}_{3/2,-3/2} = -\frac{1-\cos\theta}{2}\sin\frac{\theta}{2} \qquad d^2_{2,0} = \frac{\sqrt{6}}{4}\sin^2\theta \qquad d^2_{1,1} = \frac{1+\cos\theta}{2}(2\cos\theta - 1)$$

$$d^{3/2}_{1/2,1/2} = \frac{3\cos\theta - 1}{2}\cos\frac{\theta}{2} \qquad d^2_{2,-1} = -\frac{1-\cos\theta}{2}\sin\theta \qquad d^2_{1,0} = -\sqrt{\frac{3}{2}}\sin\theta\cos\theta$$

$$d^{3/2}_{1/2,-1/2} = -\frac{3\cos\theta + 1}{2}\sin\frac{\theta}{2} \qquad d^2_{2,-2} = \left(\frac{1-\cos\theta}{2}\right)^2 \qquad d^2_{1,-1} = \frac{1-\cos\theta}{2}(2\cos\theta + 1) \qquad d^2_{0,0} = \left(\frac{3}{2}\cos^2\theta - \frac{1}{2}\right)$$

Figure 43.1: The sign convention is that of Wigner (*Group Theory*, Academic Press, New York, 1959), also used by Condon and Shortley (*The Theory of Atomic Spectra*, Cambridge Univ. Press, New York, 1953), Rose (*Elementary Theory of Angular Momentum*, Wiley, New York, 1957), and Cohen (*Tables of the Clebsch-Gordan Coefficients*, North American Rockwell Science Center, Thousand Oaks, Calif., 1974).